

figure71 | GCS Evidence-Boundary Flight Map

Step 1-40 as a credibility-building procedure, not a flat checklist.

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PROCEDURE CLAIM

canonical contracts -> executable solver evidence -> public promotion gates -> viewer-visible diagnostics -> post-Step-40 showcase

Steps 1-13

evidence.domain

Foundation Runway

Kernel-to-viewer contracts become public evidence boundaries.

1 Kernel canonical contracts Established durable domain truth and removed dependence on legacy API shape.	2 Constraint catalog semantics Made equation semantics explicit and testable instead of being hidden inside solver code.	3 Incidence graph structure Provided deterministic structural evidence for decomposition and diagnostics.	4 Decomposition planner baseline Established local-to-global planning as a contract, not an optimization side effect.
5 Numeric task validation and assembly Made numeric engine consume structured tasks and produce assembly evidence.	6 Numeric residual/Jacobian/rank reports Created the first numeric evidence surface consumed by diagnostics.	7 Diagnostics DOF/rank/residual/status Separated structural evidence from numeric evidence and made status decisions deterministic.	8 Projection-aware gluing diagnostics Made failed local-to-global assembly explainable with stable subjects.
9 Session runtime transactions Made runtime the command and commit boundary.	10 IO adapters Made fixtures and scene persistence reproducible.	11 Viewer bridge contracts Created a public visualization boundary over runtime and diagnostics.	12 Contract tools Created deterministic support tooling for module-level and cross-module tests.
13 Cross-module quality gates			

Closed the first implementation batch with cross-boundary regression coverage.

Steps 14-18

evidence.numeric

Algorithm Lift

The placeholder solver becomes executable evidence.

14



Damped numeric local solve

Promoted numeric engine from placeholder to real iterative local solver.

15



JSON scene IO

Made scene persistence useful for real test and promotion workflows.

16



Diagnostics conflict/redundancy candidates

Turned typed placeholders into actionable diagnostic evidence.

17



Fixture corpus and golden digests

Reduced ad hoc test-local model construction and improved regression explainability.

18



CI-ready quality gates

Created one deterministic pre-push and CI-quality command.

Steps 19-27

evidence.planner

Scene-Generation Loop

Generated candidates must pass public solver gates before becoming fixtures.

19



Scene promotion public adapters

Made generated scenes pass through public solver contracts before promotion.

20



Scene-generation package split start

Began moving generator policy out of the CLI facade.

21



Topology/model split

Made graph generation helpers independently testable.

22



Validation/projection split

Clarified generated graph input/output contracts.

23



Parameterization/reporting split

Made generated scene evidence deterministic and reusable.

24



Repair policy split

Made candidate repair a structured module with testable edits.

25



Explorer and promotion orchestration split

Moved high-level scene-generation orchestration out of the CLI facade.

26



SceneGenerationStore containment

Contained scratch-store and path policy behind one adapter.

27



Promotion gate hardening

Made promotion gates more deterministic and less

dependent on stdout
parsing.

Steps 28-35

evidence.diagnostic

Rank Evidence Spine

Free/frozen rank evidence propagates through diagnostics, runtime, viewer, and promotion.

28



Numeric free-column rank evidence

Fixed numeric rank semantics for boundary-frozen tasks.

29



Architecture atlas synchronization

Made the visual architecture reflect implemented evidence paths.

30



Diagnostics rank propagation

Preserved numeric rank semantics through diagnostics.

31



Runtime/viewer rank evidence projection

Lets UI, promotion gates, and review tooling consume rank evidence without reading numeric internals.

32



Promotion gates consume rank evidence

Makes generated scene promotion prove full/free/frozen/nullity evidence.

33



Decomposition separator and SolveDAG deepening

Improves explainable local-to-global planning.

34



Boundary-aware runtime diagnostics

Makes runtime results carry structured rank/residual evidence, not only raw numeric reports.

35



Diagnostics conflict/redundancy deepening

Makes failed solves more actionable and reportable.

Steps 36-40

evidence.boundary

Evidence Closure

Robustness, corpus, viewer evidence, gates, and atlas sync close the batch.

36



Numeric robustness batch

Improves trustworthiness of dense numeric baseline reports.

37

Fixture and scene corpus expansion

Gives later algorithm work durable regression scenes.



38

Viewer and GUI evidence surface

Makes solver evidence visible to humans without parsing free-form text.



39

Quality gate hardening

Protects the Step 31-38 evidence paths by default and makes the sentinel visible in gate summaries.



40

Atlas and roadmap resynchronization

Re-closes the documentation, visualization, code, and test loop.

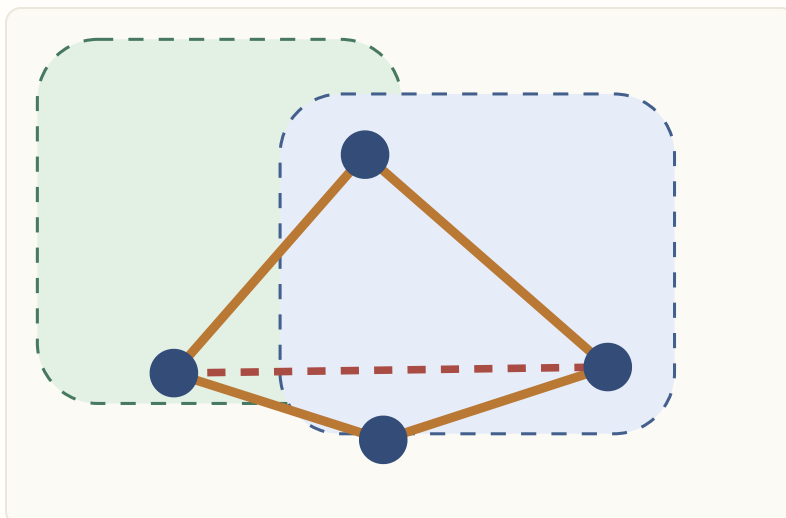


after Step 40

evidence.planner

Showcase Candidate

The next visual should combine geometry, cover, rank, gluing, and diagnostic variants.



free rank 8 / 10

frozen cols 2

gluing ok

obstruction variant